Applications and Use Cases of Machine Learning (TE7483) · ESE case-study project · AIML, Symbiosis Institute of Technology, Pune

Explainable Machine Learning Framework for Road Accident Severity Prediction and Black-Spot Detection

Dataset: UK Department for Transport STATS19, 2005-2017 (2,046,815 accidents) · Code: github.com/Manarsenic/road-accident-severity-blackspots

Abstract

Road-safety agencies must decide where to act and which conditions matter, yet most published severity models report overall accuracy on a target that is 85% one class, treat hotspot analysis separately, and use black-box models. This project builds one leak-free framework that (1) predicts accident severity (Slight, Serious, Fatal) on years the model never saw, (2) detects compact, severity-weighted black spots and validates them out of time, and (3) explains predictions with SHAP and LIME. Trained on 2006-2014, tuned on 2015 and tested once on 2016-2017, the final XGBoost model reaches macro-F1 **0.412** (majority baseline 0.301), ROC-AUC **0.698** and recalls **37%** of killed-or-seriously-injured (KSI) accidents. Accuracy is 82.1% with plain argmax and 72.3% with decision scales tuned for rare-class recall, against a 82.1% majority baseline: the headline is macro-F1 and rare-class recall because accuracy rewards ignoring fatalities. 95% of the top-100 black spots detected in 2005-14 remain active in 2015-17. A static web application runs the model, calibration and exact TreeSHAP entirely in the browser.

0.412	17.4%	0.698	95%
test macro-F1	Fatal recall	ROC-AUC	black spots persist

1. Introduction: problem, motivation, objectives

Problem. Given conditions known at the scene (road, junction, light, weather, speed limit, time, local accident history) predict whether an injury accident will be Slight, Serious or Fatal, and locate where severe accidents concentrate. **Motivation.** India recorded roughly 1.5 lakh road deaths a year (MoRTH *Road Accidents in India* reports); authorities have limited budgets and need to know where and why. The UK STATS19 data is the largest open, labelled source, so it is used for method

development; the framework transfers, a trained model would need local data. **Stakeholders.** Highway authorities, traffic police, urban planners, insurers and NGOs.

Research gap. (i) Accuracy-only reporting on an 85/14/1 target; (ii) no systematic comparison of imbalance strategies under time-based validation; (iii) prediction and hotspot analysis done separately; (iv) black-box models without explanation.

Objectives. O1 predict severity with imbalance-aware metrics validated on later years; O2 compare model families and imbalance strategies fairly; O3 detect black spots and prove they persist; O4 explain the model globally and locally and compare SHAP with LIME. **Research questions.** RQ1 Which conditions most raise the probability of Serious/Fatal outcomes? RQ2 Do black spots differ in their risk drivers? RQ3 Does leak-free spatial history improve prediction?

2. Related work

Severity prediction has been studied with logistic regression, decision trees, random forests and gradient boosting on STATS19 and similar data, typically reporting accuracy and often with random splits that mix years (Chang & Wang 2006; Abdel-Aty & Haleem 2011). Imbalance is commonly addressed by SMOTE (Chawla et al. 2002) or class weights, with mixed benefits reported. Black-spot identification traditionally uses fixed-window counting, empirical-Bayes and kernel density (Anderson 2009; Okabe et al. 2009), and density clustering such as DBSCAN (Ester et al. 1996) and HDBSCAN (Campello et al. 2013). Explainability for tree ensembles uses SHAP (Lundberg & Lee 2017) and LIME (Ribeiro et al. 2016). Gradient boosting (Chen & Guestrin 2016; Ke et al. 2017) is state of the art on tabular data, and Optuna (Akiba et al. 2019) automates tuning; isotonic calibration follows Niculescu-Mizil & Caruana (2005). This project combines these in one leak-free pipeline.

3. Dataset

Source: Department for Transport STATS19 (Open Government Licence), via the Kaggle mirror by tsiaras. File `Accident_Information.csv`: 2,047,256 rows × 34 columns, 2005-2017, one row per police-reported injury accident. Target `Accident_Severity`: 84.7% Slight, 14.0% Serious, 1.3% Fatal after cleaning.

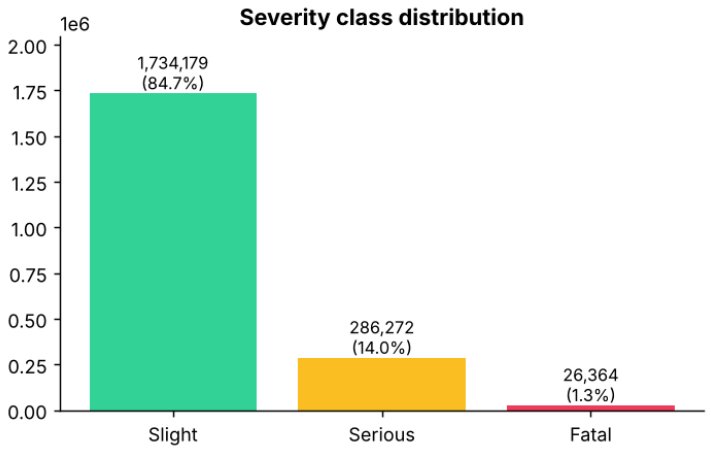


Figure 1. Class distribution of the target.

3.1 Cleaning (row counts per step)

Step	Rows remaining
raw file	2047256
drop duplicate accident ids	2047256
drop rows with missing latitude/longitude	2047081
keep coordinates inside UK bounding box	2047081
drop rows with missing time of day	2046925
keep valid speed limits (20-70 mph)	2046852
drop 'Unallocated' urban/rural rows	2046815

Only 0.022% of rows were lost. Placeholder strings ("Data missing or out of range") were converted to missing and imputed as an explicit "Unknown" level inside the sklearn pipeline (fitted on training data only). Absent hazard/special-condition codes (98% and 97% of rows) are genuine "None" levels, not missing data.

3.2 Exploratory analysis



Figure 2. Accidents by hour (area) with Serious/Fatal share (lines), and severity share by weekday. Fatal share peaks late at night, when traffic is lightest.

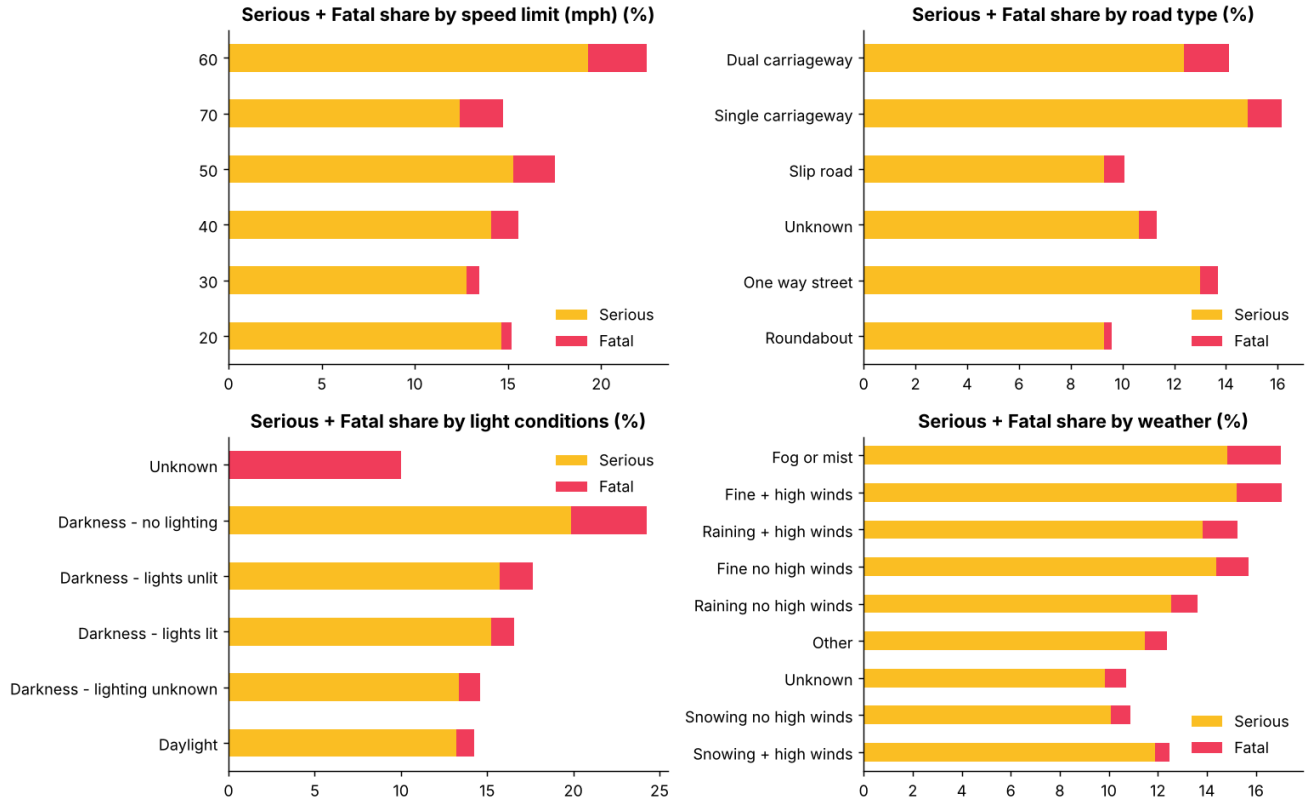


Figure 3. Serious+Fatal share by speed limit, road type, light and weather. Severity rises steeply with speed limit and in darkness without lighting; weather matters far less than expected.

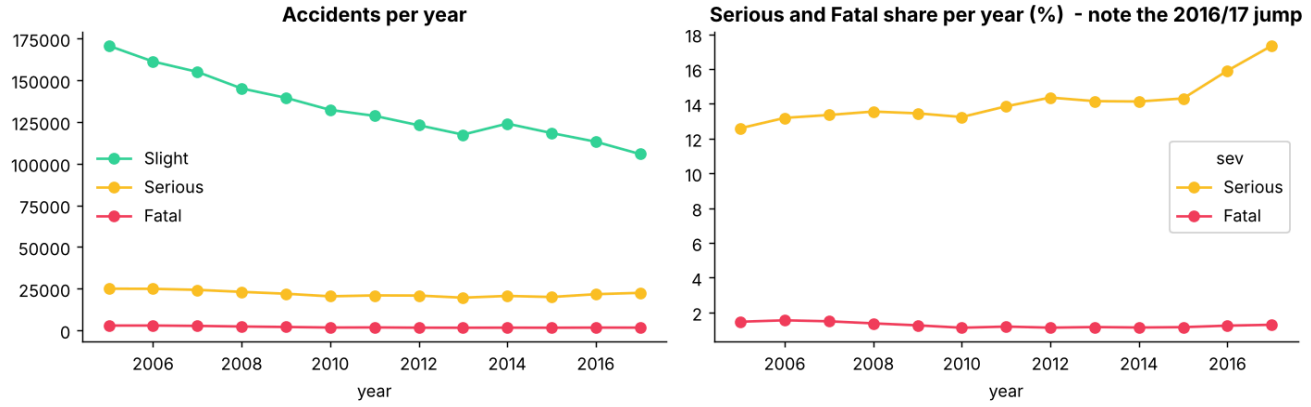


Figure 4. Accident counts fall over time while the Serious+Fatal share jumps in 2016-17 ({ksi_by_year["2015"]:.1f}% in 2015, {ksi_by_year["2017"]:.1f}% in 2017). This coincides with police forces adopting injury-based severity reporting and is a genuine distribution shift that the evaluation must face.

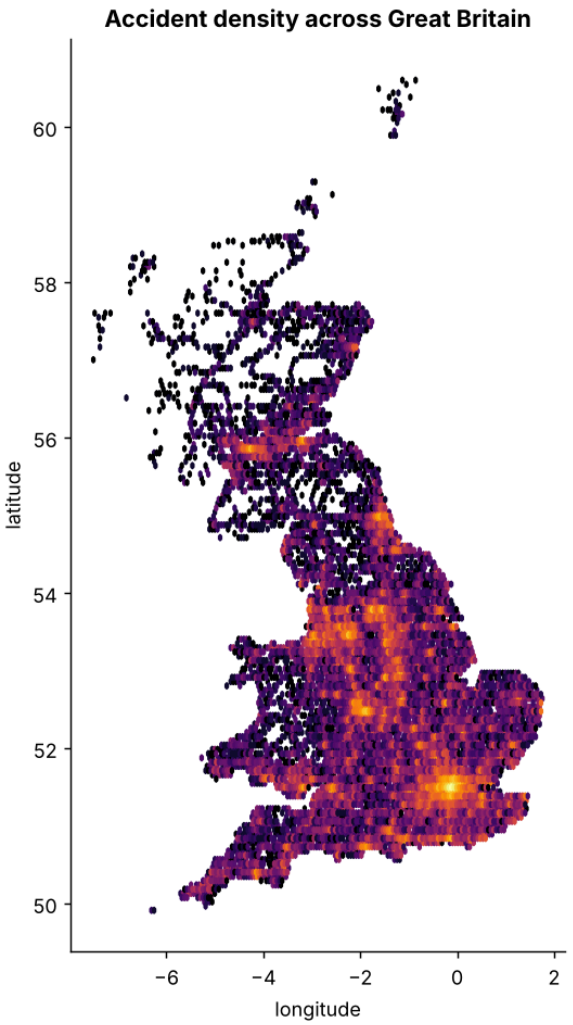


Figure 5. Accident density confirms the UK shape and concentration in cities.

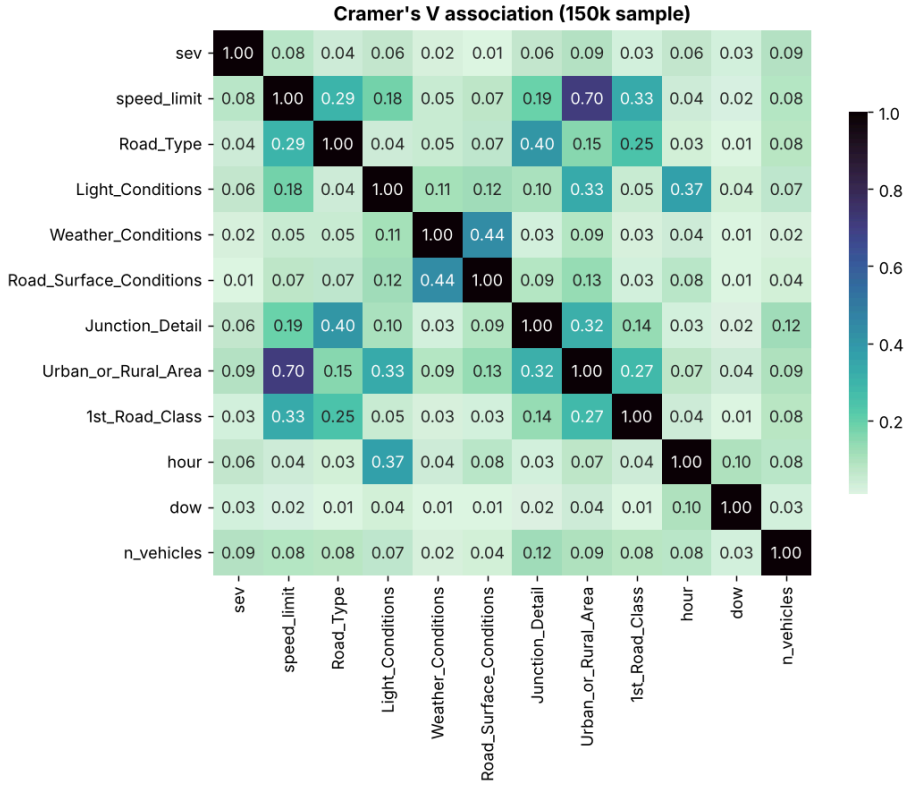


Figure 6. Cramér V association between categorical features and severity: all individual associations are weak, which foreshadows the modest ceiling on predictive performance.

3.3 Leakage control

A feature is used only if it describes the scene or is known when the accident occurs. `Number_of_Casualties` and `Did_Police_Officer_Attend_Scene_of_Accident` are consequences of severity and are excluded (Figure 7); their effect is quantified in the ablation (Section 7.4).

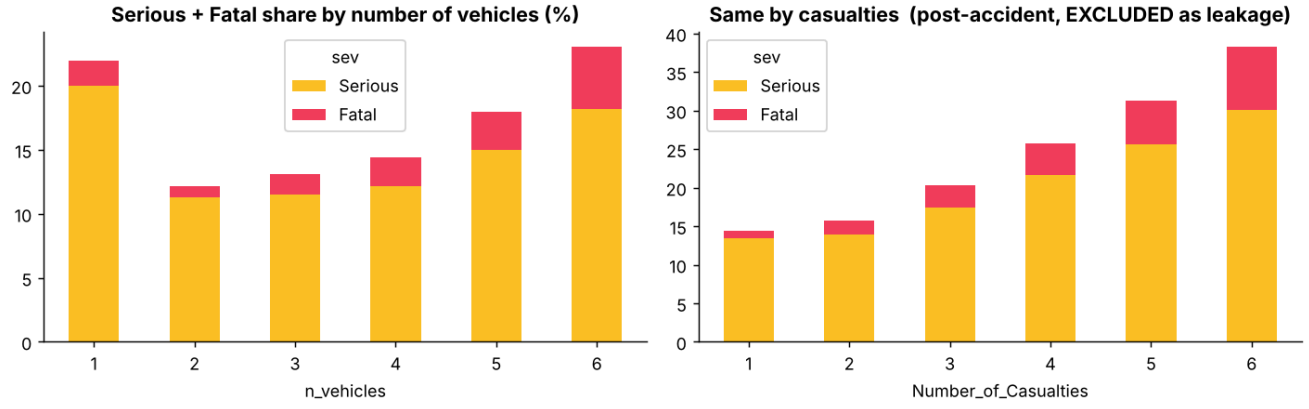


Figure 7. Fatal share rises with the number of casualties because casualties are a consequence of severity, not a predictor available beforehand.

Feature	Type	Known before outcome	Used	Reason
Number_of_Casualties	int	No	No	Consequence of the accident
Did_Police_Officer_Attend_Scene_of_Accident	cat	No	No	Recorded after the event, tied to severity

Easting/Northing, LSOA, Local_Authority_(Highway), InScotland	various	Yes	No	Redundant with latitude/longitude, district, force
Road, junction, light, weather, surface, speed limit, urban/rural	cat/int	Yes	Yes	Scene description
Date, time, weekday, month, vehicles (capped at 6)	int	Yes	Yes	Known at the time
Latitude, longitude	float	Yes	Yes	Location
h_* spatial history	float	Yes	Yes	Built only from earlier years

3.4 Feature engineering

- **Time:** hour, month, weekday, is_weekend, is_rush_hour (07-09, 16-19), is_night, season, cyclical hour (sin, cos).
- **Scene interactions:** dark_and_unlit, wet_surface, bad_weather, junction_at_night, high_speed_rural, banded speed limit (≤30, 40-50, ≥60).
- **Leak-free spatial history:** for an accident in year Y, the accident count, Bayesian-smoothed KSI rate and EPDO score of its ~550 m grid cell and its local authority, and the distance to the nearest earlier hotspot cell, all computed from years strictly before Y (expanding window), so no row ever sees its own label. Training starts in 2006 because 2005 only seeds the history.

3.5 Data dictionary

The complete data dictionary (meaning, type, example, missing %, handling) for all 56 processed columns is in `reports/tables/data_dictionary_processed.csv` and on the web app's Data page.

4. Methodology

4.1 Pipeline

> **Raw CSV** → clean (UK bounding box, valid speed limits, duplicates) → engineer scene and time features → expanding-window spatial history → **time-based split** (train 2006-14: 1,441,672 rows; validate 2015: 140,008; test 2016-17: 266,524) → sklearn `ColumnTransformer` (median impute + one-hot with rare-level grouping) fitted on training only → imbalance strategy → model → isotonic calibration and class decision scales fitted on 2015 → one-shot test evaluation → SHAP/LIME and black-spot analysis → static web app.

4.2 Splitting and validation

Primary protocol is time-based (train past, validate next year, test latest two years), which mimics deployment. A secondary stratified random split (70/15/15) is reported for comparability with the literature. Hyper-parameters are tuned with Optuna against validation-year macro-F1, and 5-fold stratified CV with paired t-tests compares the model families. The test years are never used for tuning, calibration or threshold selection.

4.3 Imbalance strategies compared

No handling (baseline); class weights (balanced; for XGBoost via sample weights); random under-sampling of Slight to 60k; SMOTE on the one-hot training matrix; and **decision-scale tuning**, in which per-class multipliers on predicted probabilities are searched on the validation year to maximise macro-F1. Resampling is inside the training pipeline only; validation and test sets keep the real class distribution.

4.4 Models and justification

A majority-class dummy shows why accuracy misleads; logistic regression and a decision tree are interpretable baselines; random forest, XGBoost and LightGBM are the main models because gradient-boosted trees dominate tabular data, handle mixed features and missing values, are fast, and support exact TreeSHAP. The best two (LightGBM, XGBoost) are tuned with Optuna (TPE) over learning rate, leaves/depth, estimators, child weight, subsampling, regularisation and a class-weight exponent.

4.5 Metrics

Macro-F1 (headline), per-class precision/recall/F1, KSI recall, ROC-AUC one-vs-rest, PR-AUC for Serious and Fatal, balanced accuracy, Matthews correlation, confusion matrices, expected calibration error, accuracy for reference, plus training and inference time.

5. Experimental setup

Python 3.13, scikit-learn 1.9.1, LightGBM 4.7.0, XGBoost 3.4.1, imbalanced-learn 0.14.2, SHAP 0.54.1, LIME 0.2.0.1, Optuna 5.0.0, hdbscan 0.8.44. Hardware: 2 CPU cores, 8 GB RAM, no GPU. `random_state=42` everywhere. The comparison grid trains on a stratified 200,000-row sample of the training years so that 17 configurations are affordable; the tuned final models are refit on all 1,441,672 training rows. Code: `src/`; notebooks `notebooks/01-05`.

6. Results: model comparison

Table 1 compares every model and imbalance strategy on the test years with decision scales tuned on 2015. XGBoost is the validation winner. **Key finding:** once the decision scales are tuned, models and imbalance strategies are within about 0.046 macro-F1 of each other (0.361-0.407 for the four main families), so the cheapest strategy (no resampling plus tuned decision scales) is as good as SMOTE or under-sampling. Without tuning, plain argmax with no handling reaches only macro-F1 0.303 because the model never predicts a fatality.

Model	Imbalance	Val macro-F1	Test macro-F1	Recall Fatal	Recall Serious	ROC-AUC	Accuracy	Fit (s)
Majority	none	0.305	0.301	0.000	0.000	0.500	0.821	4.300
Logistic Reg.	none	0.401	0.398	0.123	0.235	0.667	0.746	19.200
Logistic Reg.	class weights	0.355	0.361	0.221	0.536	0.641	0.559	29.600
Logistic Reg.	SMOTE	0.356	0.366	0.222	0.485	0.639	0.582	154.200

Model	Imbalance	Val macro-F1	Test macro-F1	Recall Fatal	Recall Serious	ROC-AUC	Accuracy	Fit (s)
Decision Tree	none	0.384	0.383	0.055	0.280	0.635	0.719	23.700
Decision Tree	class weights	0.345	0.349	0.319	0.436	0.593	0.552	22.900
Random Forest	none	0.402	0.402	0.201	0.334	0.686	0.698	108.500
Random Forest	class weights	0.385	0.392	0.100	0.501	0.654	0.638	73.200
Random Forest	under-sampling	0.402	0.402	0.199	0.274	0.684	0.722	37.400
LightGBM	none	0.406	0.405	0.146	0.286	0.685	0.730	23.600
LightGBM	class weights	0.385	0.389	0.113	0.489	0.672	0.638	24.500
LightGBM	under-sampling	0.408	0.405	0.096	0.294	0.681	0.734	102.600
LightGBM	SMOTE	0.409	0.405	0.165	0.271	0.682	0.733	205.700
XGBoost	none	0.410	0.407	0.150	0.280	0.689	0.733	21.900
XGBoost	class weights	0.389	0.390	0.111	0.492	0.673	0.639	271.900
XGBoost	under-sampling	0.410	0.407	0.155	0.271	0.685	0.736	294.700
XGBoost	SMOTE	0.407	0.402	0.143	0.219	0.688	0.755	247.400

Table 1. Test results (2016-17), tuned decision scales, models trained on a 200k sample.

6.1 Final tuned model

After Optuna tuning ({'colsample_bytree': 0.5909124836035503, 'learning_rate': 0.0311947133693891, 'max_depth': 8, 'min_child_weight': 34, 'n_estimators': 392, 'reg_lambda': 0.9835468046820036, 'subsample': 0.6849356442713105}; class-weight exponent 0.60) and refitting on 1.44M rows, the XGBoost model scores:

Decision rule	Macro-F1	Accuracy	Bal. acc.	MCC	Recall Fatal	Recall Serious	KSI recall	ROC-AUC	PR-AUC Fatal
Tuned scales (headline)	0.412	0.723	0.434	0.153	0.174	0.314	0.373	0.698	0.058
Plain argmax	0.301	0.821	0.333	0.012	0.000	0.000	0.000	0.698	0.058
Majority baseline	0.301	0.821	0.333	0.000	0	0	0	0.500	-

On accuracy. On the 2016-17 test years 82.1% of accidents are Slight, so always predicting Slight scores 82.1% accuracy while catching no fatality. Reporting accuracy alone would therefore reward the wrong behaviour. Under a stratified random split (the protocol used by most papers) the unweighted model reaches 84.6% accuracy against a 84.6% majority baseline, and on the binary KSI task accuracy is 82.1% at threshold 0.5 against 82.1% for the majority class. Accuracy above

> 90% cannot be reached honestly with scene-level features; values that high in the literature almost always come from leaked columns (casualty counts) or mixed-year splits.

6.2 Alternative protocols

Protocol	Accuracy	Majority baseline	Macro-F1	ROC-AUC
Time-based test, tuned scales	72.3%	82.1%	0.412	0.698
Time-based test, plain argmax	82.1%	82.1%	0.301	0.698
Stratified random split, unweighted	84.6%	84.6%	0.306	0.712
Stratified random split, tuned scales	75.8%	84.6%	0.415	0.709
Binary KSI vs Slight, threshold 0.5	82.1%	82.1%	0.452	0.658
Binary KSI vs Slight, tuned threshold	75.3%	82.1%	0.587	0.658

6.3 Cross-validation and significance

Model	CV macro-F1 (mean)	Std	Paired-t p vs best
Logistic Regression	0.3242	0.0028	0.0000
Decision Tree	0.3171	0.0065	0.0000
Random Forest	0.3535	0.0029	0.0000
XGBoost	0.3725	0.0029	0.0006
LightGBM	0.3813	0.0023	NaN

5-fold stratified CV on 150k rows with class-weighted models and plain argmax; paired t-tests across folds compare the best model with each competitor.

6.4 Calibration, curves and errors

Class weighting inflates raw probabilities. Isotonic calibration fitted on 2015 reduces the expected calibration error on the test years from 0.1374 to 0.0249, so the web app's probabilities can be read as probabilities. Confusion matrices, ROC/PR curves and the subgroup error analysis are on the web app's Model lab page and in the notebooks.

Subgroup	Accidents	Fatal cases	Fatal recall	KSI recall	Accuracy
All test accidents	266,524	3371	17.4%	37.3%	72.3%
Night	74,916	1388	28.8%	45.8%	67.0%
Daytime	191,608	1983	9.5%	33.4%	74.4%
Urban	175,886	1214	3.5%	27.1%	77.4%
Rural	90,638	2157	25.3%	51.2%	62.5%
Speed limit ≤ 30 mph	180,926	1217	1.0%	28.0%	76.5%
Speed limit 40-50 mph	33,032	580	14.7%	40.9%	70.1%
Speed limit ≥ 60 mph	52,566	1574	31.2%	56.9%	59.3%

At a junction	155,894	1159	3.7%	27.8%	77.1%
Not at a junction	110,630	2212	24.6%	47.7%	65.5%

6.5 Ablation study

Variant	Macro-F1	ROC-AUC	Accuracy	Recall Fatal	Recall Serious
Full model (scene + time + location + spatial history)	0.412	0.694	0.744	0.137	0.274
- spatial history features	0.407	0.688	0.751	0.094	0.263
- time features	0.409	0.687	0.742	0.134	0.274
- raw coordinates (lat/lon)	0.410	0.694	0.742	0.138	0.277
Scene only (no history, no coordinates)	0.406	0.688	0.751	0.094	0.261
+ road-level history	0.413	0.694	0.744	0.140	0.275
+ LEAKAGE: casualties & police attendance	0.422	0.701	0.749	0.149	0.291

Each row changes one thing relative to the full model, with the same tuned scales and a 400k training sample. The final row adds the two leakage columns to show the effect leakage would have had.

7. Black-spot analysis

Definition. A black spot is a compact site (150 m radius) with a high severity-weighted accident score (EPDO: Fatal 10, Serious 5, Slight 1) over the 2005-2014 detection window. **Method.** Coordinates are projected to the British National Grid. Every accident is scored by the weighted accidents within the radius, peaks are picked greedily and neighbours within twice the radius are suppressed, giving 6,000 disjoint sites. DBSCAN and HDBSCAN were tested first and rejected as the primary method because on road networks they chain along corridors (Table 5): even with eps = 100 m the largest DBSCAN cluster holds 24,693 accidents, and HDBSCAN clusters span up to 166 km, which are not black spots. DBSCAN and a weighted KDE (100 m grid) are kept as cross-checks: 100% and 100% of the top-50 sites coincide with a KDE hotspot and a DBSCAN cluster respectively.

Method	Parameters	Clusters	Noise share	Median span (km)	P90 span (km)	Max span (km)
DBSCAN	eps=100 m, min_samples=25	6142	0.74	0.30	0.55	14.37
DBSCAN	eps=150 m, min_samples=30	5139	0.66	0.50	0.93	38.47
DBSCAN	eps=200 m, min_samples=40	3553	0.64	0.72	1.35	42.95
DBSCAN	eps=300 m, min_samples=60	2119	0.59	1.19	2.44	52.84
HDBSCAN (leaf)	min_cluster=30, min_samples=15, 500k sample	3589	0.50	1.86	8.19	165.71

Table 5. Why raw density clustering was not used as the primary definition.

7.1 Temporal validation

Sites detected on 2005-2014 were checked against 2015-2017, which the detector never saw. A site "persists" if it still averages at least one accident per year and at least half its earlier annual rate. **95%** of the top 100 persist (median later/earlier rate 0.93); a rolling check (detect 2005-11, validate 2012-14) gives 97%. As a control, ordinary dense sites around rank 1,000 persist 85% of the time, so the persistence criterion is lenient and the rank correlation between detection score and later score (Spearman 0.63 at the final setting) is the more informative evidence. The top-100 sites hold 1.3% of all accidents but have a Serious+Fatal share of 14.2%, slightly below the overall 14.8%: the busiest sites are urban junctions with many slight accidents.

7.2 Sensitivity analysis

Radius (m)	Weights F/S/SI	Top-100 KSI share	Persistence	Median later/earlier	Spearman
100	10/5/1 (final)	0.141	0.960	0.893	0.536
150	10/5/1 (final)	0.142	0.950	0.928	0.633
200	10/5/1 (final)	0.141	0.980	0.963	0.711
300	10/5/1 (final)	0.139	0.980	0.979	0.706
150	5/2/1	0.130	0.950	0.921	0.590
150	20/5/1	0.141	0.950	0.928	0.647
150	1/1/1 (counts only)	0.125	0.960	0.943	0.595

Radius (100-300 m) and severity weights (10/5/1, 5/2/1, 20/5/1, counts only) change the ranking only mildly: persistence stays 95%-98%. A radius of 150 m and the common EPDO weights 10/5/1 were chosen because they match typical UK site definitions and gave a stable ranking; the share of the final top-20 sites reappearing under other settings ranges from 60% to 100%.

Local authority	Accidents	Fatal	Serious	KSI share	EPDO score	2015-17 per yr	Persists
Westminster	343	1	58	0.17	584	31.00	True
Westminster	347	3	49	0.15	570	31.67	True
Southwark	374	1	41	0.11	547	33.33	True
Brighton and Hove	295	1	43	0.15	476	27.67	True
Westminster	296	3	38	0.14	475	25.67	True
Lambeth	295	4	35	0.13	471	32.67	True
Westminster	259	2	44	0.18	453	25.33	True
Westminster	278	1	41	0.15	451	22.00	True
Camden	244	6	36	0.17	442	23.00	True
City of London	261	3	35	0.15	428	19.00	True
Westminster	250	3	37	0.16	425	24.33	True
Brighton and Hove	230	3	40	0.19	417	12.00	True

Table 7. The 12 highest-scoring black spots. A black spot shows **concentration, not cause**, and the data has no traffic volumes, so it cannot rank roads by per-vehicle risk.

8. Explainability findings

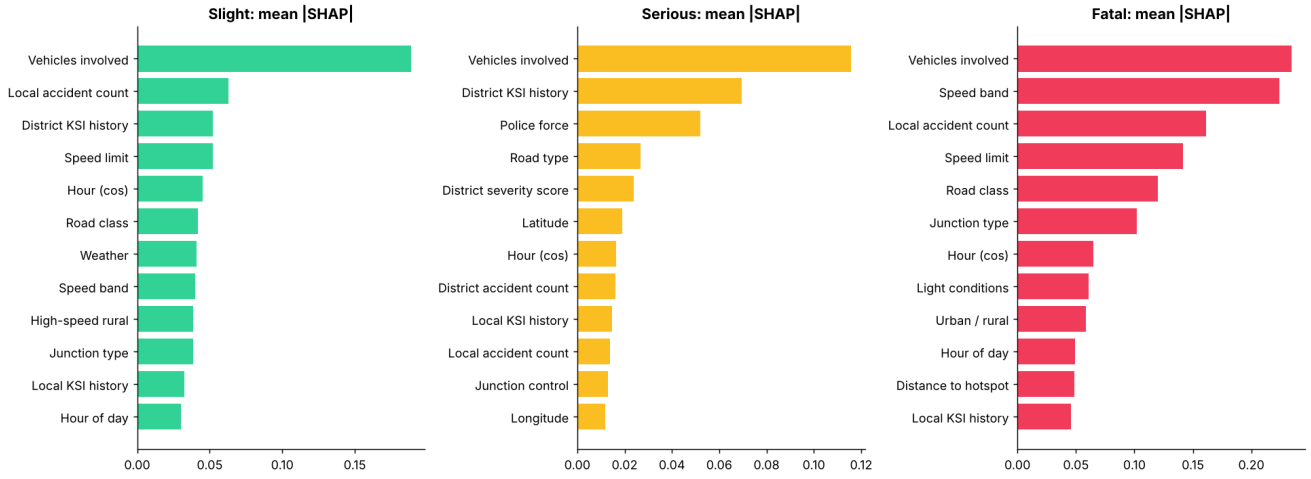


Figure 8. Mean |SHAP| per class (6,400-row test sample). Speed limit, local accident history and the hour dominate; weather is minor.

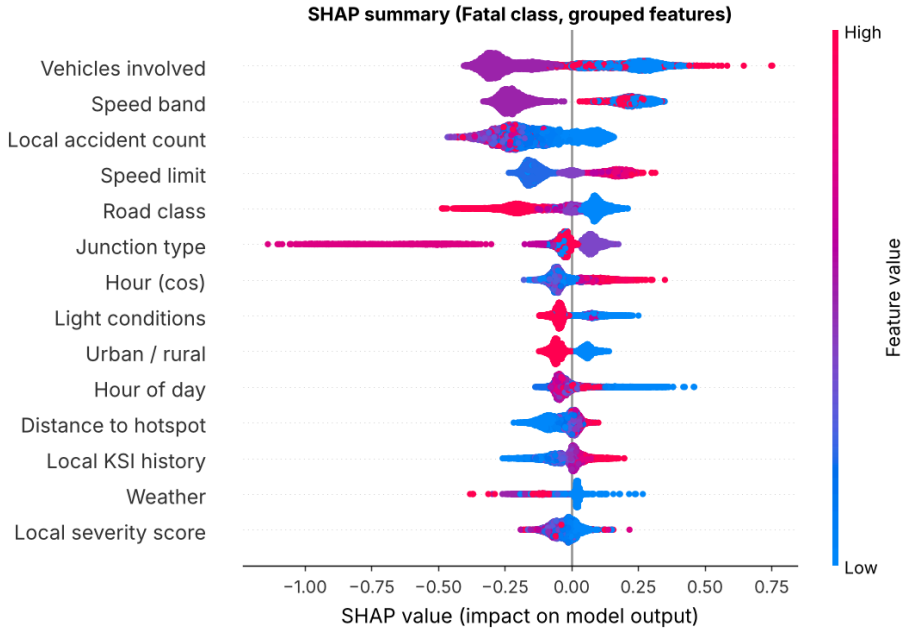


Figure 9. SHAP summary for the Fatal class.

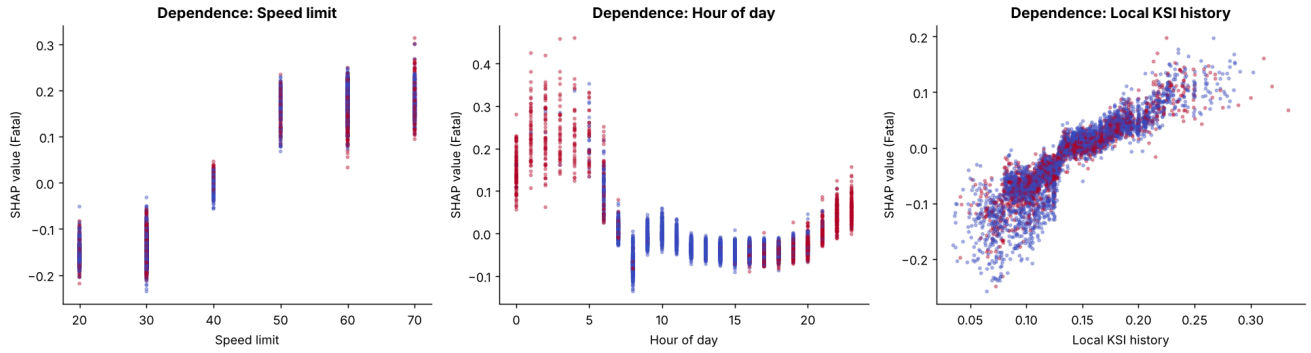


Figure 10. Dependence plots: SHAP value for Fatal versus speed limit, hour and local KSI history (colour: night).

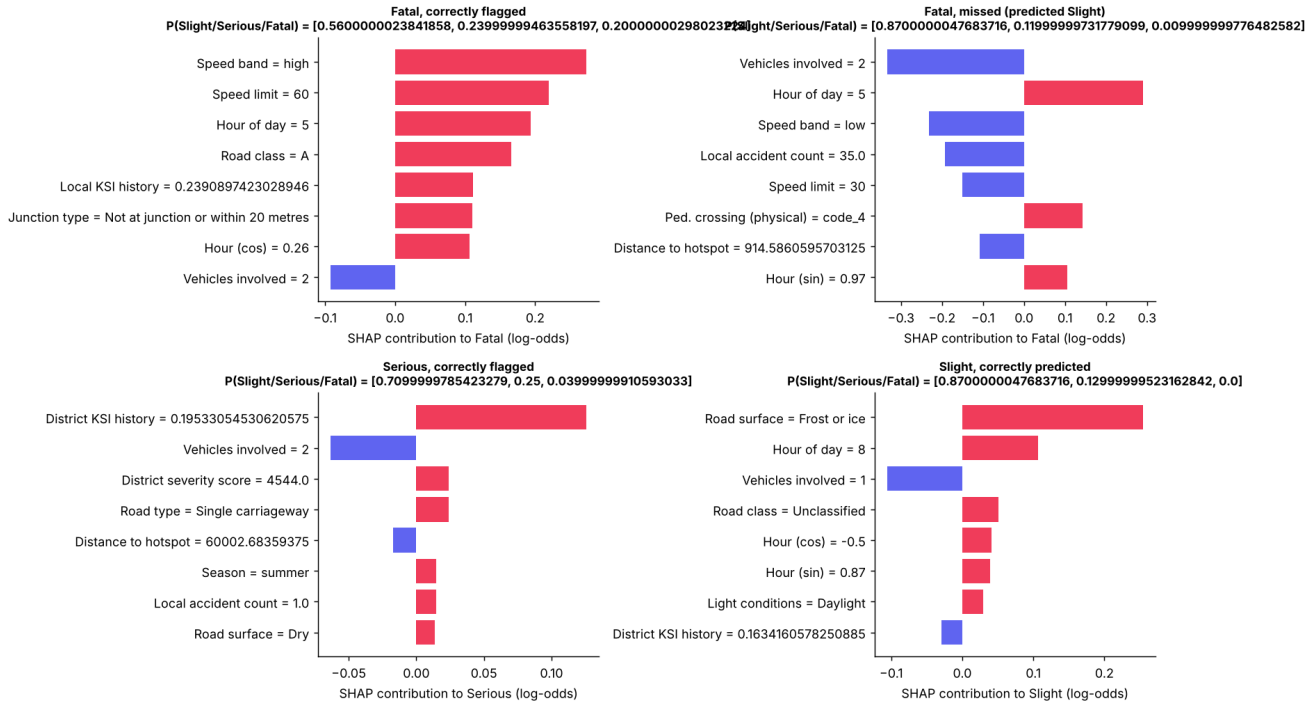


Figure 11. Local explanations: a correctly flagged fatal, a missed fatal, a correctly flagged serious and a correctly predicted slight accident.

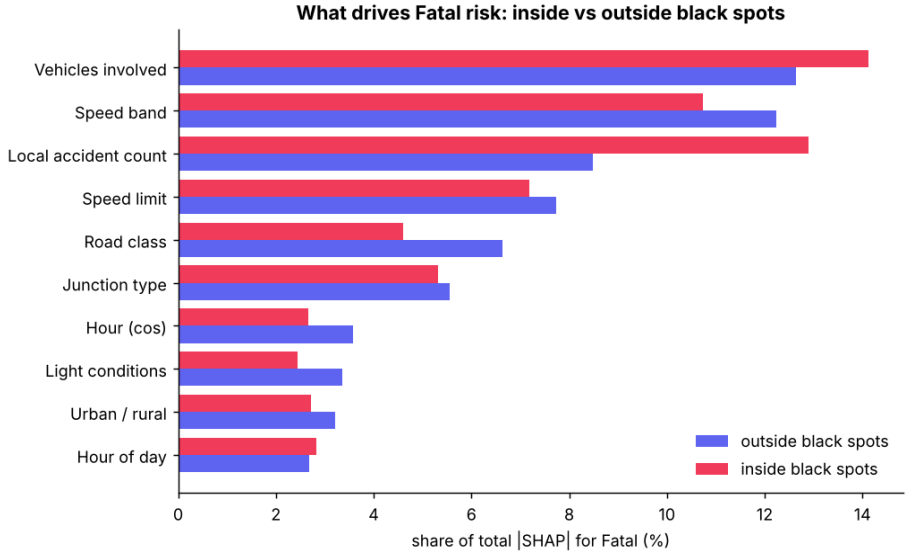


Figure 12. Share of total |SHAP| per feature inside versus outside black spots.

SHAP vs LIME. SHAP is exact and deterministic for trees; LIME fits a random local surrogate. The two agree on the main drivers for some cases and differ for others; LIME's top-5 features also vary with the random seed.

Class	SHAP top-5	LIME top-5	Overlap	LIME stability (Jaccard)
Fatal	Speed band, Speed limit, Hour of day, Road class, Local KSI history	Local authority, Vehicles involved, Speed limit, Light conditions, Speed band	0.40	0.55
Serious	District KSI history, Vehicles involved, District severity score, Road type, Distance to hotspot	District KSI history, Vehicles involved, Local authority, Carriageway hazards, Weather	0.40	0.61

Class	SHAP top-5	LIME top-5	Overlap	LIME stability (Jaccard)
Slight	Road surface, Hour of day, Vehicles involved, Road class, Hour (cos)	Vehicles involved, Carriageway hazards, Weather, Road type, Local authority	0.20	0.38

Recommendations (association, not causation). Speed management on high-limit rural roads, lighting on unlit stretches, and junction treatment at the top-ranked urban sites are the levers the model associates with lower severity; any intervention must be evaluated with before-after studies.

9. Application

The web application (React, fully static, deployable on GitHub Pages) has six pages: Overview; Predictor (map-based location, scenario inputs, calibrated probabilities, live exact TreeSHAP and what-if analysis, all computed in the browser from the exported trees); Black-spot map (heatmap, markers, filters, ranked table, CSV export); Explainability; Model lab; and Data & method. A parity test (`app/scripts/parity.mjs`) confirms the JavaScript engine reproduces scikit-learn probabilities and SHAP values. A Streamlit version (`streamlit_app/`) uses the full Python pipeline.

10. Limitations and ethical considerations

- UK-only, injury accidents only: slight accidents are under-reported, uninjured collisions are absent.
- Scene-level features cap performance (ROC-AUC 0.698); vehicle type, speeds, driver behaviour and casualty details are not in this table. Joining the vehicle and casualty tables is the clearest route to better prediction.
- No traffic exposure: black spots show concentration, not risk per vehicle.
- A reporting-system change around 2016 creates a distribution shift; we handle it with calibration and scales fitted on 2015, but residual shift remains.
- SHAP and LIME explain the model, not causes. Predictions support planning and must not be used to blame individuals; locations are accident sites, not people.

11. Conclusion and future work

A leak-free, time-validated framework links severity prediction, black-spot detection and explanation. The central lesson is methodological: with an 85/14/1 target, accuracy hides failure, and once decision scales are tuned the choice of model and imbalance method matters little next to feature information. Future work: join vehicle and casualty tables; add Indian MoRTH/state data and test transfer; normalise by traffic exposure; road-network (graph) clustering; spatio-temporal forecasting of next-year hotspots; live-data deployment.

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